Topics

- Limits of Sequences
- Algebraic operations + limits in $\mathbb{R}$
- Limits in $\mathbb{R}^d$

Sequences + Limits

We can think of $\mathbb{N}$ as space or time.

Dynamic perspective

Statement holds eventually if there exists $N \in \mathbb{N}$

s.t. Statement is true for all $n \geq N$.

Ex • $\frac{1}{2^n}$ is eventually less than 0.0001.

• $2^n + (-1)^n 1000$ is eventually positive.

Static perspective

Statement holds for all but finitely many $n \in \mathbb{N}$.

Lemma Suppose $A \subset \mathbb{N}$. TFAE:

(a) $\exists N$ s.t. $n \geq N \Rightarrow n \in A$ (b) $n \in A$ for all but finitely many n.

$\Updownarrow$ $\Updownarrow$

(c) A^c is bounded $\Leftrightarrow$ (d) A^c is finite

Def A sequence (p_n) in a metric space X

Converges if there exists $p \in X$ s.t. for all $\varepsilon > 0$

$\exists N \in \mathbb{N}$ s.t. $n \geq N$ implies $d(p_n, p) < \varepsilon$. We write

$\lim_{n \to \infty} p_n = p$, or $p_n \to p$. If (p_n) does not

converge, it **diverges**.

Example:

- $p_n = \frac{1}{n}$ converges to 0 in $\mathbb{R}$
- $p_n = n$ diverges in $\mathbb{R}$
- $p_n = \frac{1}{n}$ diverges in $\mathbb{R}_{>0}$

Unless specified, consider sequences in $\mathbb{R}$ with std. metric.

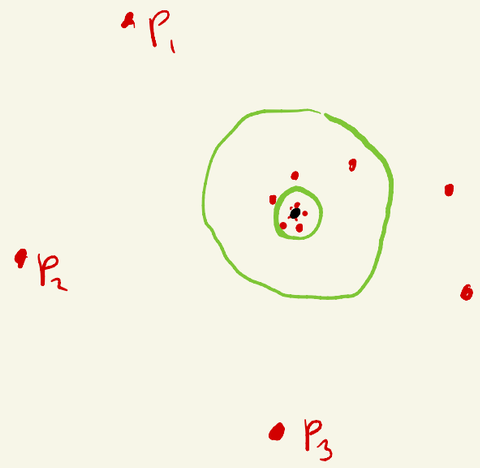
Def The **image** or **range** of (p_n) is the set $\{p_1, p_2, p_3, \dots\}$. A seq. is **bounded** if it has bounded image.

Examples

	bounded?	finite image?	convergent?
$p_n = \frac{1}{n}$	yes	no	yes
$p_n = n^2$	no	no	no
$p_n = (-1)^n$	yes	yes	no
$p_n = 1$	yes	yes	yes

Static perspective on convergence

p_n converges to p iff for all $\varepsilon > 0$, $N_\varepsilon(p)$ contains p_n for all but finitely many n.



Lemma (p_n) converges to p in a metric space X iff $d(p_n, p)$ converges to 0 in $\mathbb{R}$.

Proof: $d_{\mathbb{R}}(d(p_n,p), 0) = |d(p_n,p) - 0| = d(p_n,p)$, so the definitions are saying the same thing.

Thm If $p_n \to p$ and $p_n \to p'$ for $p, p' \in X$,

then $p = p'$.

Proof

Let $\varepsilon > 0$. $\exists\ N, N'$ s.t.

$$n \geq N \Rightarrow d(p_n, p) < \varepsilon/2$$

$$n \geq N' \Rightarrow d(p_n, p') < \varepsilon/2$$

So $n \geq \max(N, N')$ implies

$$d(p, p') \leq d(p, p_n) + d(p', p_n) < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

Thm 3.2.c

(p_n) converges $\Rightarrow$ (p_n) bounded

Proof $\exists N$ s.t. $n \geq N \Rightarrow d(p, p_n) < 1$.

$\{p_1, \ldots, p_{N-1}\}$ is finite, so bounded by $R = \max\{d(p, p_n) \mid 1 \leq n < N\}$.

We have $d(p, p_n) \leq \max\{R, 1\}$ for all $n \in \mathbb{N}$.

Thm 3.2.d

If $E \subset X$, and $p \in E'$, then

there is a sequence (p_n) in E s.t. $p = \lim_{n \to \infty} p_n$.

proof For each n, we can choose $p_n \in E$ s.t. $d(p_n, p) < \frac{1}{n}$.

The seq. (p_n) converges to p.

Thm (33) Suppose (s_n), (t_n) sequences in $\mathbb{R}$ s.t. $s_n \to s$ and $t_n \to t$.

(a) $\lim_{n\to\infty} (s_n + t_n) = s + t$

(b) $\lim_{n\to\infty} c s_n = cs$, $\lim_{n\to\infty} (s_n + c) = s + c$ for $c \in \mathbb{R}$.

(c) $\lim_{n\to\infty} (s_n t_n) = st$

(d) $\lim \frac{1}{s_n} = \frac{1}{s}$ if $s_n \neq 0$ and $s \neq 0$

Proof

(a) Let $\varepsilon > 0$. $\exists\, N_1, N_2$ s.t.

$n \geq N_1 \Rightarrow |s_n - s| < \frac{\varepsilon}{2}$

$n \geq N_2 \Rightarrow |t_n - t| < \frac{\varepsilon}{2}$

To get (*), write $s_n = s + (s_n - s)$, $t_n = t + (t_n - s)$

$$s_n t_n = \big(s + (s_n - s)\big)\big(t + (t_n - t)\big)$$
$$= st + s(t_n - t) + t(s_n - s) + (s_n - s)(t_n - t)$$

(b) Exercise

(c) Use the identity

(*) $s_n t_n - st = (s_n - s)(t_n - t) + s(t_n - t) + t(s_n - s)$

Let $\varepsilon > 0$. $\exists\ N_1, N_2$ s.t.

$n \geq N_1 \Rightarrow |s_n - s| < \sqrt{\varepsilon}$

$n \geq N_2 \Rightarrow |t_n - t| < \sqrt{\varepsilon}$

So $n \geq \max\{N_1, N_2\}$ implies

$|(s_n - s)(t_n - t)| < \varepsilon$

So $(s_n - s)(t_n - t) \longrightarrow 0$.

Using (a), (b), and (*) $s_n t_n - st \longrightarrow 0$.

(d) Exercise

Lemma (Case of Squeeze theorem) Suppose a_n, b_n real seqs.

If $0 \le a_n \le b_n$ and $b_n \to 0$, then $a_n \to 0$.

Proof: Let $\varepsilon > 0$. If $n \ge N$ implies $|b_n| < \varepsilon$,
then it also implies $|a_n| < \varepsilon$, because $|a_n| \le |b_n|$.

Thm (3.4a)

A seq $\{X_n\}$ in $\mathbb{R}^d$ converges to $X \in \mathbb{R}^d$

iff $X_{j,n} \longrightarrow X_j$ for $j = 1, \dots, d$ where $X_n = (X_{1,n}, \dots, X_{d,n})$

and $X = (X_1, \dots, X_d)$.

Proof Suppose $X_n \longrightarrow X$. $|X_{j,n} - X_j| \leq |X_n - X|$, so $X_{j,n} \longrightarrow X_j \ \forall j$.

Suppose $X_{j,n} \longrightarrow X_j \ \forall j$. Let $\varepsilon > 0$.

$\exists N_1, \dots, N_d$ s.t. $n \geq N_j \Rightarrow |X_{j,n} - X_j| < \frac{\varepsilon}{\sqrt{d}}$

Let $N = \max\{N_1, \dots, N_d\}$

$$n \geq N \Rightarrow |X_n - X| = \sqrt{\sum_{j=1}^{d} (X_{j,n} - X_j)^2} < \sqrt{\sum_{j=1}^{d} \left(\frac{\varepsilon}{\sqrt{d}}\right)^2} = \varepsilon$$

Thm (3.46) Let $\{\vec{x}_n\}$, $\{\vec{y}_n\}$ be seqs in $\mathbb{R}^k$, $\{\beta_n\}$ seq in $\mathbb{R}$ s.t.
$\vec{x}_n \to x$, $\vec{y}_n \to y$, $\beta_n \to \beta$.

(a) $\lim x_n + y_n = x + y$

(b) $\lim x_n \cdot y_n = x \cdot y$

(c) $\lim \beta_n x_n = \beta x$